Identifying Persistent Fecal-Indicator Pollution Hotspots for Water Rehabilitation Planning: A Multi-Source, Multi-Region Water Quality Data Pipeline

Charlene Labilles
Mapua University

Antonio Paulo Cuyo
Mapua University

Alfonso Vicente
Mapua University

***Abstract*—Water-quality monitoring data is collected by many organizations in different formats, vocabularies and units, so it is hard to answer a simple planning question: which places have pollution problems that keep coming back? We built a reproducible data pipeline that ingests fecal-indicator observations from three monitoring programmes (the US Water Quality Portal, GEMStat and the European Environment Agency's Eionet, the last two through the Open Water Quality compilation), daily weather from Open-Meteo and administrative boundaries from Natural Earth. The pipeline harmonizes 1.36 million raw observation rows into 25,108 observations at 300 sampled sites, stores them in a PostgreSQL warehouse, validates every stage with automated checks, and is orchestrated by one Apache Airflow DAG that runs in Docker. From the warehouse we flag persistent hotspots, rank administrative regions for rehabilitation priority, and test whether exceedance is higher after rain. The results cover the United States, Europe and Mexico only: Africa has no data in the extract and Asia has no *E. coli* rows, so the results are not global comparisons. At the sampled sites, exceedance is rare in Europe (about 3% of samples), common in the United States (14–15%) and very common in Mexico (56%), and it is higher on wet days in the United States and Mexico. We explain the design decisions and trade-offs behind each layer, the evidence that the pipeline is correct and reproducible, and the limits of what the results can support.**

***Index Terms*—data engineering, ETL pipeline, Apache Airflow, PostgreSQL, data quality validation, water quality, fecal indicator bacteria, reproducibility**

I. Introduction

A. Problem Statement

Water pollution data is usually obtained from many different monitoring stations and organizations, which makes it hard to consolidate and analyze this data as well as find the locations with recurring water pollution problems. The main goal of this research is to create an effective data pipeline for integrating large water quality data with environmental data like precipitation and temperature to detect pollution hot spots.

Whether a water source is considered safe is usually decided by counting fecal indicator bacteria: Escherichia coli in fresh water and intestinal enterococci in marine water.

Unfortunately the data is fragmented, thus planners have difficulty answering which places have pollution problems that keep coming back, year after year? A single high reading may be a one-off event. A site that exceeds safety thresholds in most years is a structural problem that needs rehabilitation.

This project is a data-engineering answer to that problem. Its product is not a one-time analysis but a pipeline: every step from download to dashboard is code, configuration and automated checks, so that the result can be regenerated, audited and extended.

Code, configuration, tests and all evidence are in the project repository (https://github.com/labillescharlene21-lab/project); the results are published as a static dashboard (https://labillescharlene21-lab.github.io/project/).

B. Objectives

1) Automatically ingest fecal indicator, weather, and boundary data from 5 independent sources in multiple formats. Harmonize observations into one schema, one indicator vocabulary, and one unit (CFU/100 mL).
2) Apply automated data-quality checks at the raw, staging, and curated layers.
3) Store the curated data model in PostgreSQL with keys and relationships.
4) Orchestrate the full flow with Apache Airflow in a Dockerized environment.
5) Identify persistent hotspot sites and rank regions for rehabilitation priority. Measure whether rainfall is associated with higher exceedance rates at hotspot sites

C. Key questions and stakeholders

The data product answers four questions: (1) Which monitoring sites exceed the single-sample safety threshold repeatedly across years; (2) Which administrative regions contain the most persistent hotspots; (3) Are exceedance rates higher after rainfall than in dry conditions; and (4) Which regions should be prioritized for rehabilitation, and why. The primary users are water-rehabilitation planners and river-basin organizations; regulators, public-health agencies, local governments and researchers use the same curated data for compliance follow-up, recreational-risk screening and reuse.

Final technical report, October 2026.

D. Scope and Limitations

The study period is 2015–2025. Only *E. coli* (fresh water) and enterococci (marine water) are scored, against the single-sample statistical threshold values of the US EPA 2012 Recreational Water Quality Criteria: 410 CFU/100 mL and 130 CFU/100 mL respectively [1]. Fecal and total coliform are ingested and reported but never scored.

The researchers configured a stratified sample over five strata: the United States, Europe, and three GEMStat regions (Africa, Asia and Latin America). **Only three strata produced scored sites: the United States, Europe and Mexico.** Africa has no rows at all in the GEMStat extract we obtained, and the only Asian country in the extract (India) has no *E. coli* measurements, so no Asian site qualifies for fresh-water scoring (Section XI-A). Every result in this report therefore describes the United States, Europe and Mexico. It is *not* a global comparison, and the dashboard says so next to the ranking. Groundwater, drinking-water compliance, chemical pollutants, real-time alerting and data before 2015 are out of scope.

Decision: sample sites instead of ingesting everything. We take up to $K = 60$ sites per stratum and water realm, chosen by a seeded random draw among sites with at least 24 samples of their primary indicator in at least three distinct years. **Why.** Open data is concentrated in the United States and Europe; equal-sized strata stop well-documented regions from dominating a ranking, keep the weather download feasible (Section XII) and keep the pipeline quick to rerun. A fixed seed makes the same configuration select the same sites. The raw volume (1.36 million rows) still far exceeds what a small sample would give. **Cost.** Results describe the sampled sites, not every water body.

II. Data Sources

A. The five sources

Table I lists what the final run ingested. Two of the five sources, GEMStat and Eionet, are distributed together through the Open Water Quality (OWQ) compilation, which harmonizes nine public monitoring programmes into one export.

B. Why these sources, and what each choice cost

Decision: take GEMStat and Eionet from the OWQ compilation, not from their native APIs. Why. One scripted export returns two programmes in a single layout, with locations, a harmonized indicator name and a unit already attached. Writing and maintaining separate clients for each national or regional programme was not feasible in a seven-day project. **Cost.** Provenance is one step removed: the compilation is itself a research product (cited as in preparation), it has its own cut-off dates, it fills censored values before export so censoring is invisible for these sources, and its `region` column mixes in countries we did not ask for (Canada and the United States appear in the GEMStat extract). We handle these in staging rather than hiding them, and native APIs are listed under future work.

Decision: for the Water Quality Portal, fetch the site catalogue first, sample, and only then request results. Why. The site summary (one row per site, indicator and year, 540,786 rows) is small enough to download and already carries result counts and years, so eligibility (at least 24 samples in at least three years) can be decided *before* any result is requested. We then sample 120 sites and request results only for those, which cuts the download to 13,240 rows. A single national summary request did time out, so the catalogue is fetched in smaller requests (58 files). **Cost.** The WQP sample is fixed before any value is seen, so the sites are drawn from the eligibility rule rather than from the data, and site types outside our realm mapping (about 10,455 sites, such as BEACH-programme sites, canals and the Great Lakes) are never considered.

Decision: use gridded reanalysis weather (Open-Meteo) at 0.25° cells, not station data. Why. Global coverage with one API, no keys, and a consistent daily series for every site in every country. Sites share cells, so 300 sites need only 256 requests. **Cost.** Reanalysis is a modelled field, not a rain gauge at the site; a cell is about 25 km wide; and the free tier limits the pull to roughly 156 cells per day, which stretched the first download over several days.

Decision: assign sites to administrative regions (Natural Earth admin-1), not river basins. Why. Administrative regions have clean public boundaries and match the level at which regulators and local governments act. A site more than 25 km from any region is left unassigned; a site just outside a polygon is matched to the nearest region and flagged. **Cost.** Rehabilitation is planned by watershed, not by province. Basin-based aggregation is a listed future improvement.

C. What profiling found before any cleaning

We profiled every source on the raw layer before writing transformation rules, so the rules respond to observed problems. Table II lists the findings that changed a design decision. The full profile is in the repository (`docs/source_profiling.md`).

D. Terms of use

GEMStat's open tier is equivalent to CC BY 4.0 and EEA data is CC BY 4.0; Open-Meteo is CC BY 4.0 for non-commercial use; the Water Quality Portal is US public data with a requested citation. The OWQ header names no open licence: observations remain the property of the contributing programmes and must be cited to them directly, which we do (see References). The project is non-commercial, so all terms are compatible. The attribution header of every raw file is kept in the repository as evidence. Sources: Open Water Quality [2], GEMStat [3], the European Environment Agency [4], the Water Quality Portal [5], Open-Meteo [6] (whose archive is based on the ERA5 reanalysis [7]) and Natural Earth [8].

TABLE I
SOURCES, ACCESS METHODS AND FINAL-RUN VOLUMES (RAW LAYER).

Source	Access	Raw format	Role and final-run volume
OWQ: GEMStat (UNEP GEMS/Water)	Scripted HTTP export	CSV with # header	Latin America stratum. 263,711 rows, 5,691 sites, 2015–2023; five countries only
OWQ: Eionet (European Environment Agency)	Scripted HTTP export	CSV with # header	Europe stratum. 1,085,694 rows, 27,390 sites, 2015–2021; 29 regions
Water Quality Portal (USGS, EPA, NWQMC)	REST API: site summary, then results	CSV	United States stratum. 540,786 summary rows over 130,918 sites; results fetched only for the 120 sampled sites (13,240 rows)
Open-Meteo archive	REST API, one request per grid cell	JSON	Daily rainfall and temperature. 256 cells × 4,018 days = 1,028,608 rows
Natural Earth admin-1	File download	Shapefile	Region boundaries. 4,596 features, 251 countries

TABLE II
RAW-DATA FINDINGS THAT SHAPED THE PIPELINE.

Finding	Consequence
GEMStat has five regions only (Mexico 168,254 rows, India 86,489, Uruguay 6,531, Canada 2,024, United States 413); no African rows	The Africa stratum is empty. Strata are assigned from the spatially joined country, not from the `region` text
India has fecal coliform (54,858), total coliform (27,272) and enterococci (4,359) rows, but no *E. coli*	No Asian freshwater site meets the eligibility rule (2,067 candidate sites, 0 eligible). Reported as a finding, not repaired
Eionet water body "other" is 65% of rows	Excluded: it cannot be classified as fresh or marine, and the wrong threshold would be applied. Costs 18,060 sites
Eionet ends 2021-12-21; GEMStat thins out after 2020	Each area's "most recent five years" are different calendar years. Stated as a limitation
GEMStat values cluster at 24,000, 24,196 and 2,419,600	These look like laboratory reporting maxima that are not flagged as censored. Severity is understated; exceedance flags are unaffected
WQP results mix units and censoring (`<10`, `*Non-detect`, units `MPN`, `hours`, `None`)	A mapping table, a censoring rule, and explicit drop reasons for rows that cannot be converted
WQP summary has year values from 1 to 2206	Rows outside 2015–2025 are dropped when site counts are built
Open-Meteo: 0% missing precipitation or temperature over 1,028,608 cell-days	No gap-filling needed; weather coverage is complete
1 of 4,596 Natural Earth polygons is invalid	Not repaired; affects only sites inside it. Stated as a limitation

III. ARCHITECTURE

A. Layered design

The pipeline moves data through three layers, with an automated validation gate after each one (Fig. 1). Each layer has a strict rule about what may happen to the data in it (Table III).

Decision: keep raw data immutable and put every judgement in a later, documented layer. Why. If a rule turns out to be wrong (for example the half-detection-limit substitution), only staging or curated has to be rebuilt; nothing has to be downloaded again, and a checksum proves the raw files did not change. This also gives every database row a lineage back to a raw file and batch (`raw_batch_id`, `raw_file`). **Cost.** Extra disk (about 0.4 GB of raw files) and a second pass over the data to apply rules.

TABLE III
LAYER CONTRACTS.

Layer	Allowed	Not allowed
Raw	Save files exactly as received; add a manifest (source, request, timestamp, row count, SHA-256 checksum)	Renaming columns, filtering rows, fixing values
Staging	Type casting, unit and name standardization, censored-value handling, deduplication, scope filtering, site sampling, spatial join	Aggregation, business scoring
Curated	Joins across sources, derived metrics, hotspot and priority rules	Changing staging values without a documented rule

B. Technology choices and trade-offs

Table IV gives the reason for each tool and what we gave up. The dominant constraint was scale: after sampling, the working data is 25,108 observations, so a single-machine stack is the right size.

The tools are Apache Airflow [9], PostgreSQL [10], Apache Parquet [11] and Docker Compose [12].

C. Design principles

Four principles run through every layer: *configuration over code* (thresholds, weights, sampling and unit mappings live in YAML files, so changing a rule changes the result without editing Python); *determinism* (fixed seed, natural keys, fixed tie-breaks, so reruns give identical output); *idempotency* (every step can be rerun, and the load is an upsert); and *fail early on critical problems, record the rest* (critical validation failures stop the task, warnings are stored but do not).

IV. IMPLEMENTATION

A. Code organization

The repository contains about 5,900 lines of Python in the pipeline code and 254 automated tests. The code is organized by pipeline stage so that each member's work

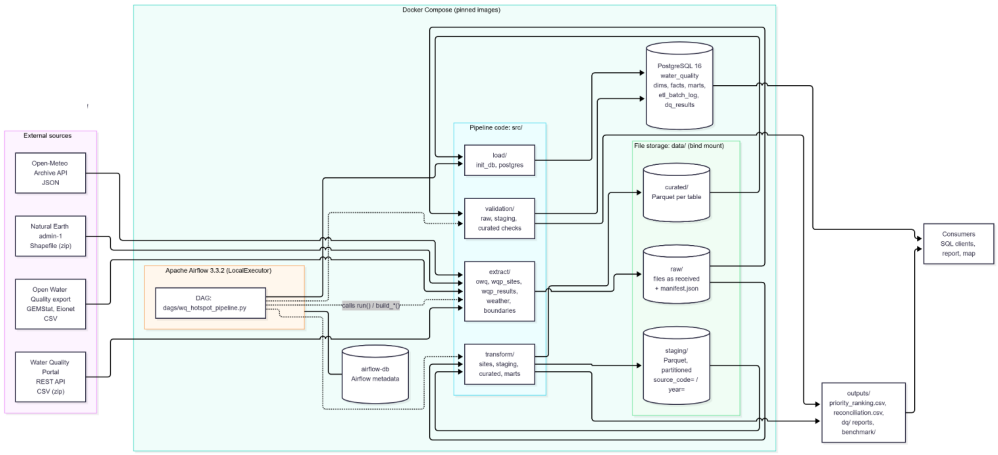


Fig. 1. Data Architecture. Solid arrows move data; dashed arrows are validation gates. Critical failures stop the run.

TABLE IV
TECHNOLOGY CHOICES.

Component	Tool	Why, and the trade-off
Processing	pandas, PyArrow, GeoPandas	The sampled data fits comfortably in memory; GeoPandas does the spatial join. A distributed engine such as Spark would add operational weight for no benefit at this size, but would be needed if we ingested every site
Staging format	Parquet, partitioned by `source_code= /year=`	Columnar, compressed and typed; partitions let a reader open only the slice it needs (Section X). Less convenient to open by hand than CSV
Warehouse	PostgreSQL 16	Keys, foreign keys, `CHECK` constraints and `ON CONFLICT` upserts give rerun safety and integrity. A columnar warehouse would be faster for huge scans but is unnecessary here
Orchestration	Apache Airflow 3	Task dependencies, retries, parameters, scheduling and per-task logs. A shell script or cron job would be simpler but gives no retries, no run history and no failure diagnostics
Environment	Docker Compose	Same services and versions on every machine; the cost is Docker's memory and disk needs (at least 4 GB memory)
Language	Python 3.11	Mature libraries for HTTP, tabular and geospatial work

has a clear boundary: `src/extract` (one module per source), `src/transform` (sites, harmonization, staging, curated layer, marts), `src/validation` (one module per stage), `src/load` (schema creation and the PostgreSQL load), `src/analysis` (EDA, profiling, benchmark, dashboard export), `src/utils` (HTTP, manifests, paths, sampling) and `dags` (the Airflow DAG). All rules and parameters are in `config/`: sources, sampling, business rules, unit and indicator mappings, and one schema file per source and layer. The DAG file contains no business logic; each task calls a function in `src`, so every step can also be run and tested from the command line.

B. Extraction

Each extractor writes its files and a `manifest.json` into an immutable batch folder. The manifest records the source, the request parameters, the time, every file with its row count and SHA-256 checksum, and a status (`success`, `partial` or `failed`). Three behaviours matter for reliability:

- **Resilient HTTP.** One shared client applies a 120 s timeout and three retries with exponential back-off (2, 4 and 8 seconds) on statuses 429, 500, 502, 503 and 504. This was necessary in practice: a national WQP summary request timed out, and Open-Meteo enforces a daily quota.
- **Resumable downloads.** The weather extractor skips every grid cell that is already complete. If the daily quota runs out, the batch is marked `partial`, raw validation fails, and the same trigger can be repeated the next day to continue.
- **A snapshot mode.** The default `PIPELINE_MODE=snapshot` blocks every network call and reuses the files already in `data/raw/`; `live` downloads from the sources. The default is deliberately the safe one, so a rerun can never change the input by accident.

C. Stratified sampling

Sampling is part of the pipeline, not a manual step. A site is *eligible* if it has at least 24 samples of its primary indicator in at least three distinct years. From the eligible sites, up to 60 per stratum and realm (fresh or marine) are drawn with a fixed random seed (42). Strata are defined in `config/sampling.yaml` by source and by the country or continent a site falls in (from the spatial join); for example, the United States and Canada are excluded from the Latin America stratum. The Water Quality Portal sample is drawn from its site catalogue before its results are requested; the OWQ sample is drawn by the `build_sampled_sites` task, after all three site sources

TABLE V
WAREHOUSE TABLES AND FINAL-RUN ROW COUNTS.

Table	Grain (primary key)	Final run
dim_source	one row per source	3
dim_indicator	one row per indicator	4 (2 scored, 2 supplementary)
dim_region	one row per admin-1 region	107
dim_grid_cell	one row per 0.25° cell	256
dim_site	one row per sampled site	300
fact_observation	one row per observation	25,108
fact_weather_daily	cell and day	1,028,608
mart_site_hotspot	site and indicator	285
mart_region_priority	one row per region	97
etl_batch_log, dq_results	one row per load task or validation check	grows with each run

are extracted. Weather is requested only for the grid cells that contain a sampled site, which is why the weather task depends on sampling in the DAG (Section VIII).

D. Testing

Tests use small synthetic datasets written with the same writers and readers the pipeline uses, and a mocked HTTP layer (`responses`), so they run in seconds without network access. They cover the extraction rules, the harmonization edge cases (censored values, unit mapping, replicate collapsing), the hotspot and priority arithmetic against hand-computed answers, determinism (a rerun gives identical output), every validation check on passing and failing input, and the report generators. Of the 254 tests, 11 need the PostgreSQL service and run inside the Docker environment. Writing the tests first caught real bugs; for example, a test of a helper script showed that it resolved paths against the wrong folder when pointed at another README.

V. DATA MODEL

The warehouse is a small dimensional model: five dimensions, two fact tables, two mart tables holding the analytical results, and two operational tables that log loads and validation results.

A. Design decisions

Decision: natural keys, not surrogate keys. What. `site_key` combines the source code and the source's own site identifier, `obs_key` the source code and the record identifier, a region is its Natural Earth code and a cell is its coordinates. **Why.** Keys built from the data itself are identical on every rerun, which is what makes the load an idempotent upsert (`INSERT...ONCONFLICTDOUPDATE`) and makes a rerun with unchanged input a no-op. **Cost.** Keys are longer text values than integers, and a source that changed its own identifiers would create new rows. At 25,108 observations the extra storage is irrelevant.

Decision: copy the weather onto each observation, and keep the daily table as the source of truth. Why. Every analysis question asks about rain *at the time of an observation*. Storing `rain_48h_mm`, `rain_72h_mm` and `is_wet` on the fact row makes those questions a single-table query. **Cost.** Redundancy: the same rainfall appears on many observation rows. It stays consistent because the copy is derived by the pipeline, not typed in, and `fact_weather_daily` can recompute it for any other window.

Decision: store supplementary indicators, but never score them. Why. Fecal and total coliform are useful context (and were most of the Asian data), but no current recreational criterion exists for them. They are kept in `fact_observation` with `is_scored=false` so nothing is thrown away, and the marts ignore them.

Decision: precompute the analytical results as tables, not views. Why. The persistence rule, the Theil–Sen trend and the medians are easier to write and test in Python than in SQL, and a stored mart is what the dashboard export and the ranking CSV read. The marts are loaded by the same upsert as every other table. **Cost.** A rule change needs the marts rebuilt, which the DAG does on every run.

Integrity is enforced in the database, not only in code: primary and foreign keys, `CHECK` constraints (for example `value_cfu_100ml>=0`) and lineage columns. The curated validation stage (Section VII) reads the foreign keys from the database catalogue, so a new foreign key is checked automatically.

VI. TRANSFORMATIONS AND BUSINESS RULES

A. Staging: harmonization

The staging step turns three differently shaped sources into one observation table. In order, it: casts types; maps each source's indicator names to four codes (`e_coli`, `enterococci`, `fecal_coliform`, `total_coliform`); converts units to CFU/100 mL; parses censored values; assigns the realm (fresh or marine) from the water-body type; removes quality-control blanks and non-water media; deduplicates; keeps only the sampled sites; and spatially joins each site to an administrative region. Anything that cannot be converted is *dropped with a recorded reason*, not silently, so the layers can be reconciled (Section VII).

Decision: treat MPN as CFU, and substitute half the detection limit for left-censored values. What. Results in MPN/100 mL count as CFU/100 mL (factor 1.0). A value "$< x$" counts as $x/2$ and "$> x$" as x. **Why.** OWQ makes the same two assumptions, so WQP and OWQ values are treated identically and the sources stay comparable. Both estimate viable bacteria per 100 mL. Because right-censored values are underestimates, the rule can only understate exceedance. **Alternatives.** Survival-analysis methods for censored data (listed as future work), or dropping censored rows (which would bias the data towards high values). **Cost.** The assumptions are stated as limitations, not hidden; 3.8% of staged values are censored.

B. Curated layer: rainfall, exceedance and sample-days

The curated layer joins each observation to the weather of its site's grid cell and applies the first business rules.

- **Antecedent rain.** Rain on the sample day plus the previous day (48 h), and one more day (72 h). A sample-day is *wet* if the 48 h rain is at least 10 mm. Observations with missing weather are excluded from both the wet and the dry rates, because counting unknown weather as dry would understate the wet effect.
- **Sample-days.** Several samples at one site on one day count as one sample-day, using a shifted geometric mean $v = \exp\left(\frac{1}{n}\sum_i \ln(v_i + 1)\right) - 1$. EPA summarizes bacteria with geometric means because counts are approximately log-normal; the $+1$ shift allows zero values and is negligible at the hundreds of CFU/100 mL that matter.
- **Exceedance.** A sample-day exceeds when v is above the realm's threshold (410 for *E. coli*, 130 for enterococci).

C. Marts: persistence, trend and priority

A site-year is *classified* if it has at least five sample-days, and *poor* if more than 10% of its sample-days exceed. The 10% comes from the way EPA defines its threshold: it should not be exceeded in more than 10% of samples. A site is a *persistent hotspot* if it was poor in at least three of its most recent five classified years. The trend is the Theil–Sen slope [13] of the annual exceedance rate (needs three or more years).

Regions are ranked by a weighted score of four components, each min–max normalized across ranked regions:

$$S_r = 0.40\,\tilde{P}_r + 0.30\,\tilde{V}_r + 0.15\,\tilde{T}_r + 0.15\,\tilde{W}_r, \qquad (1)$$

where P_r is the share of the region's sites that are persistent hotspots, V_r the median over its sites of the median ratio to the threshold (severity), T_r the median site trend, and W_r the median ratio of wet to dry exceedance rate. A region needs at least three sampled sites to be ranked; ties are broken by number of hotspots, then region code, so ranks are identical on every rerun.

Decision: adapt EPA's rule to calendar years, and say so. EPA applies its criteria to 30-day windows; most of our sites are sampled a few times a month at best, so a 30-day window is nearly always empty. We apply the same 10% rule per calendar year with a five-sample-day minimum. With fewer than five days, one exceedance (20% or more) would make the year poor on its own. **Cost.** The rule is an adaptation of a published standard, not the standard itself, and it is stated as such.

Table VI lists the other rules, the reason for each, and its status. Every value is a setting in `config/business_rules.yaml`; the document `docs/business_rules.md` labels each as a *standard* (cited), a *data* decision, or an *assumption*.

VII. Data Quality and Validation

A. Approach

Validation runs automatically at three points, each with its own checks, and every check writes a pass/fail result as JSON and as a row in the `dq_results` table. A *critical* failure stops the Airflow task and so everything downstream; a *warning* is recorded and the run continues. Table VII summarizes the checks. The full catalogue is in Appendix B.

Decision: validate at the layer boundaries, and make most checks critical. Why. A bad file caught at the raw boundary costs seconds; the same problem found in a dashboard costs trust. The checks that protect correctness (checksums, keys, referential integrity, reconciliation) are critical so that a bad run *cannot* reach the warehouse unnoticed. **Where we chose warnings.** Conditions that are *findings*, not corruption: an empty stratum, an implausibly large value, weather coverage below 90%, a suspicious hotspot share. They must be visible but should not block a run that is otherwise correct. **Cost.** A warning can be ignored; that is why the two warnings of the final run are explained below and reported on the dashboard.

B. The final run

All 42 checks ran in the final run (Table VIII): no critical failure, and two warnings. The warnings are expected and informative: `stratum_coverage` fails for the five stratum-and-realm combinations that have no sampled site (Africa and Asia in both realms, and Latin America marine); this is the check that records the coverage limitation of Section XI-A. `value_range` flags values above 10^6 CFU/100 mL, which in this data are the 2,419,600 laboratory upper limit; they are kept and flagged, not removed.

C. Reconciliation: every row is accounted for

The strongest single check is the reconciliation: for each source, raw rows minus the rows dropped with a recorded reason must equal the staging rows, and the staging rows must equal the curated rows and the rows in `fact_observation`. It holds for every source (Table IX). Almost all dropped OWQ rows are simply sites that were not sampled, by design; the WQP drops are real data-quality losses and are listed by reason.

VIII. Orchestration

One Airflow DAG, `wq_hotspot_pipeline`, runs every step from extraction to curated validation (Fig. 2). Each task calls a function in `src` and returns the path of its manifest or batch reference, which Airflow passes to the next task, so a task always reads the batch its upstream task produced.

A. Design decisions

Decision: one DAG, with validation as ordinary tasks in the dependency chain. Why. Because `validate_raw` and `validate_staging` sit between the build tasks, a critical failure marks everything downstream *upstream failed*, so bad data cannot reach staging, curated or the warehouse. Validation is therefore a gate, not a report written afterwards. `build_marts` and `load_postgres` come before `validate_curated` because the curated checks run against the loaded database. **Alternative.** A chain of scripts run by hand. We rejected it: it has no

TABLE VI
BUSINESS RULES AND THEIR BASIS.

Rule	Value	Basis and trade-off
Safety thresholds	410 and 130 CFU/100 mL	Standard: EPA 2012 single-sample values [1]
Persistence window	last 5 classified years	Assumption: the goal is *current* problems; old ones may be fixed
Persistent hotspot	poor in ≥ 3 of 5 years	Assumption: a majority of recent years; one or two poor years can be a single storm season
Wet day	≥ 10 mm in 48 h	Assumption: no universal standard; moderate rain that typically produces runoff
Minimum sites to rank	3	Assumption: with one or two sites a region's persistence share can only take two or three values, so a single site would decide the score
Weights	0.40 / 0.30 / 0.15 / 0.15	Assumption: persistence is our definition of a hotspot, so it weighs most
Sites per stratum and realm	60, seed 42	Data: keeps strata balanced and the weather pull feasible
Eionet "other" water bodies	excluded	Data: cannot be classified as fresh or marine

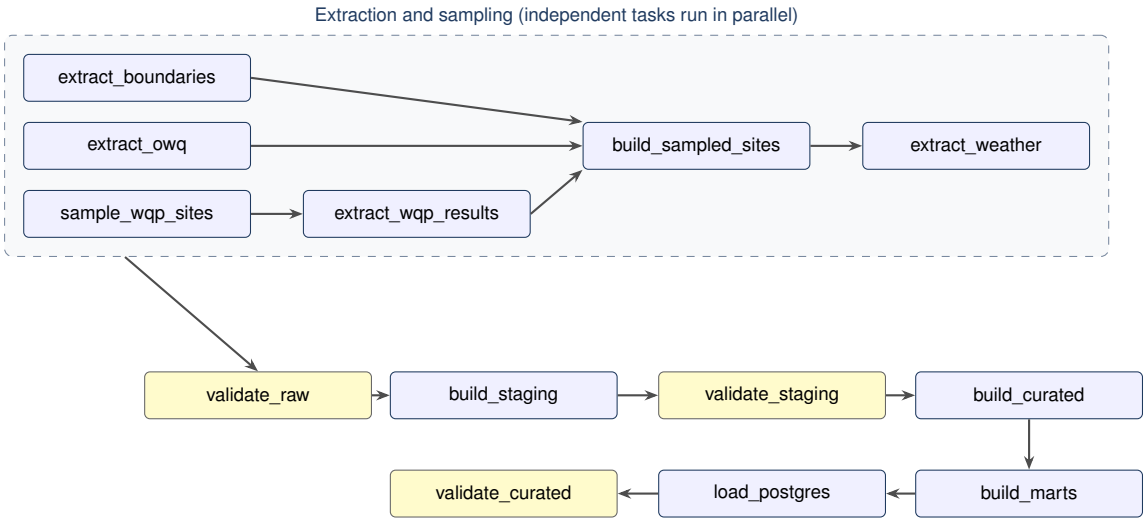


Fig. 2. The DAG. Yellow tasks are validation gates. The weather download depends on sampling, because only the grid cells that contain a sampled site are requested.

TABLE VII
VALIDATION STAGES AND WHAT EACH GUARDS AGAINST.

Stage	When	Guards against
Raw (VAL-1)	after extraction, before staging reads	incomplete downloads (a `partial` manifest), missing or empty files, files changed after download (SHA-256), row-count drift, missing columns, malformed weather series
Staging (VAL-2)	after staging is built	wrong types, null keys, duplicate observation keys, unaccepted codes, negative or implausible values, dates outside the period, invalid coordinates, orphan observations, empty strata
Curated (VAL-3)	after the load, against PostgreSQL	missing tables, orphan foreign keys, duplicated natural keys, row counts that do not reconcile across layers, weather coverage below 90%, inconsistent ranks, a hotspot share of 0% or 100%

TABLE VIII
VALIDATION RESULTS OF THE FINAL RUN (FROM THE WAREHOUSE, `DOCS/EVIDENCE/FINAL_RUN_EVIDENCE.TXT`).

Stage	Checks	Passed	Crit. fail	Warn.
Raw	26	26	0	0
Staging	9	7	0	2
Curated	7	7	0	0

TABLE IX
ROW RECONCILIATION ACROSS LAYERS (FINAL RUN).

Source	Raw	Dropped	Staging	Warehouse
WQP	13,240	1,873	11,367	11,367
OWQ GEMStat	263,711	259,100	4,611	4,611
OWQ Eionet	1,085,694	1,076,564	9,130	9,130
Total	**1,362,645**	**1,337,537**	**25,108**	**25,108**

WQP drops: unmapped unit 1,449; non-water medium 273; QC blank 108; non-numeric value 31; rejected result 12.

retries, no history of what ran, and nothing stops a later script from running after an earlier one failed.

Decision: retry transient failures, but fail immediately on errors a retry cannot fix. What. Tasks retry twice, with a delay of two minutes that grows to at most ten. `ConfigError` (for example a start year of 1800) and `DataQualityError` (a critical validation failure) fail the task at once. **Why.** A timeout or rate limit often clears on its own, and a retry is cheap. A bad parameter or a corrupt file will not fix itself, and waiting through retries only delays the diagnosis.

a) Parameters and scheduling.: The DAG takes four parameters, `start_year`, `end_year`, `k` and `seed`; anything not given comes from `config/sampling.yaml`. The schedule is `@monthly` with `catchup=False` and `max_active_runs=1`, so runs never overlap and the DAG never replays history.

b) Failure diagnostics.: When a task fails, a callback writes one log line with the DAG, task, run, try number, batch identifier and exception, so a failure can be traced to the exact raw batch without reading each log. A failed run is kept as evidence (`docs/evidence/dag_run_failure.txt`).

c) Idempotency.: We loaded the same curated data twice more after the DAG's own load. Each load inserted zero rows, rewrote the 1,054,768 existing rows with identical values, and left every table's row count unchanged (`docs/evidence/rerun_idempotency_final.txt`). This is what the upsert on natural keys is for: a rerun is safe.

d) A lesson about scheduling.: Unpausing a monthly DAG with `catchup=False` makes Airflow create one run for the latest interval immediately. Our first instructions also told the reader to trigger a run, which started a second one behind it. Because the pipeline is idempotent the second run changes nothing, but it is exactly the kind of surprise that leaves two sets of run records in the logs and makes evidence look inconsistent. The README now says to unpause only.

IX. Deployment and Reproducibility

A. Environment

Docker Compose starts the whole stack: a PostgreSQL 16.15 warehouse; a second PostgreSQL 16.15 for Airflow's own metadata; the Airflow 3.3.2 API server, scheduler and DAG processor; a one-time initialization service; and the pipeline image, built from the official Airflow image with our Python dependencies pinned in `requirements.txt`. The services have health checks, the project folders are mounted from the host (so code changes need no rebuild and outputs appear on the host), and named volumes keep the warehouse, the Airflow database and the Airflow logs.

Decision: pin every version, and make the safe mode the default. Why. The pipeline must give the same answer on a teammate's laptop and an evaluator's. Pinned images and libraries remove version drift; the default `snapshot` mode removes input drift, because it cannot download anything. **Cost.** A first run on an empty machine must set `PIPELINE_MODE=live` and then takes days, because of the weather quota; we do not publish the 0.4 GB of raw files. Publishing them is the main remaining reproducibility gap (Section XIV).

B. Evidence of reproducibility

We tested reproducibility the way an evaluator would. After the final run on the development machine, we cloned the branch into an empty folder, copied in the raw files (which are not in the repository), kept the default settings, built the stack, ran the DAG once and queried the new warehouse. The result was identical: 25,108 observations, 1,028,608 weather rows, 285 scored sites and 97 regions, with all 42 validation checks giving the same outcome (`docs/evidence/fresh_clone_evidence.txt`). With the raw files present, the whole DAG finished in a few minutes.

TABLE X
Format comparison on the 1,349,405 raw OWQ rows (nine columns). Size in MB; write, full read and two-column read times in seconds.

Format	MB	Write s	Read s	2 cols s
CSV	100.2	16.1	1.75	1.20
JSON Lines	257.0	11.4	19.5	15.2
Parquet (snappy)	6.9	1.1	0.21	0.06

C. Consuming the results

The marts are queried with plain SQL (`sql/queries.sql`) or exported as `priority_ranking.csv`. A static dashboard, published with GitHub Pages, shows a map of every monitored site coloured by severity, the top ten regions and sites, and the scope notice. Its data file is generated from the curated layer by `src/analysis/export_site.py`, including the coverage statement, so the page cannot claim more than the data holds.

X. Formats and Partitioning

We measured the choice of formats rather than assuming it (`docs/format_benchmark.md`, reproducible with one command). The benchmark writes and reads the pipeline's own data in CSV, JSON Lines and Parquet: the 25,108 staged observations, and all 1,349,405 raw OWQ rows. It records file size, write time, read time, the time to read two columns, and whether column types survive the round trip. Times are medians of three runs in the Docker container.

Parquet is about 15 times smaller than CSV and 37 times smaller than JSON, reads the whole file roughly 8 and 90 times faster, and reading two columns is around 20 times faster than CSV because only those columns are read. On the staged observations the pattern is the same, and only Parquet keeps types such as dates exactly. We also timed a read filtered to one partition (`source_code=wqp` and `year=2025`) against a full scan of the partitioned staging dataset: the filtered read opened 1 of 25 files and 911 of 25,108 rows, and was roughly ten times faster, although both take milliseconds at this size. The point is that read cost follows the slice requested, not the whole history.

Decision: source-faithful formats in raw, Parquet in staging and curated, CSV only where people open files. Raw keeps what each provider delivers (CSV, JSON, shapefile) because raw must be source-faithful. Everything that the pipeline reads repeatedly uses typed, partitioned Parquet. CSV remains for small hand-off files and for the bulk `COPY` into PostgreSQL. The partition key, `source_code` and `year`, is how every downstream step already slices the data.

XI. Results

A. What the sample covers

Table XI shows the outcome of the sampling design. Three strata produced sites; two did not, for reasons we could trace to the data rather than to the pipeline.

TABLE XI
SAMPLING OUTCOME PER STRATUM AND REALM.

Stratum	Realm	Sampled	Note
United States (WQP)	fresh, marine	60 + 60	includes Puerto Rico (5) and US Virgin Islands (3)
Europe (Eionet)	fresh, marine	60 + 60	13 countries; Spain 47 and Germany 36 are 69% of the 120 sites
Latin America (GEMStat)	fresh	60	all 60 sites are in Mexico; no marine sites
Asia (GEMStat)	fresh	0	2,067 candidate sites, none eligible: India has no *E. coli* rows
Africa (GEMStat)	—	0	no African rows in the extract

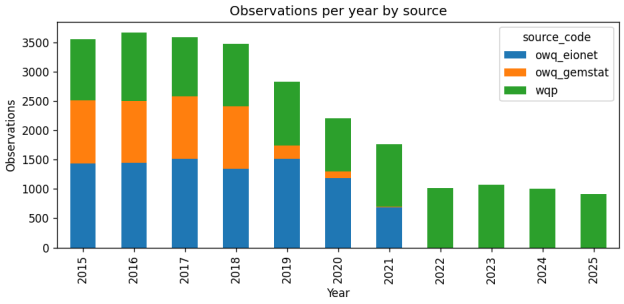


Fig. 3. Staged observations per year and source (25,108 in total).

TABLE XII
VALUES AND EXCEEDANCE OF THE SCORED INDICATORS (STAGED OBSERVATIONS). MEDIAN IN CFU/100 ML; "ABOVE" IS THE SHARE OF SAMPLES ABOVE THE THRESHOLD.

Source	Indicator	Obs.	Median	Above
Eionet	*E. coli*	4,571	15	2.9%
Eionet	enterococci	4,559	10	3.4%
GEMStat	*E. coli*	1,556	750	55.6%
WQP	*E. coli*	3,884	41	13.7%
WQP	enterococci	4,672	12	15.4%

Of the 300 sampled sites, 285 have scored data; the other 15 are WQP sites with no usable observation after harmonization. Table IX shows where the rows went.

B. Observations over time

Coverage over time is very uneven (Fig. 3). The Water Quality Portal covers the whole period, Eionet ends in 2021 and the sampled GEMStat sites are effectively limited to 2015–2018. Because each site is judged on its most recent five classified years, each stratum is judged on different calendar years. We state this wherever strata are compared.

C. Exceedance and persistent hotspots

Table XII gives the share of samples above the threshold; Table XIII the share of sites that are persistent hotspots. Both show the same ordering: European bathing sites are rarely above the threshold, the United States sits in between, and the Mexican river stations exceed it most of the time.

These contrasts must not be read as the water being "cleaner" in one place. The strata measure different things: Mexican river stations, European bathing waters and mixed US networks, over different years. Also, 21.9% of GEMStat *E. coli* values (34.2% of fecal coliform and 49.4% of total coliform) sit at a laboratory upper reporting limit, meaning

TABLE XIII
PERSISTENT HOTSPOTS BY STRATUM AND REALM (88 OF 285 SCORED SITES).

Stratum	Realm	Sites	Hotspots	Share
Europe	fresh	60	3	5%
Europe	marine	60	2	3%
Mexico (GEMStat)	fresh	60	53	88%
United States	fresh	52	16	31%
United States	marine	53	14	26%

TABLE XIV
WET VERSUS DRY EXCEEDANCE RATE PER SITE (MEDIAN OVER SITES WITH BOTH CONDITIONS). THE LAST COLUMN IS THE SHARE OF SITES WHOSE WET RATE IS HIGHER.

Stratum	Realm	Sites	Wet	Dry	Wet>dry
Europe	fresh	54	0.00	0.00	13%
Europe	marine	54	0.00	0.00	20%
Mexico (GEMStat)	fresh	59	0.80	0.52	73%
United States	fresh	50	0.25	0.05	68%
United States	marine	53	0.14	0.04	66%
All	all	270	0.09	0.04	48%

"at least this much", so severity for Mexico is if anything understated.

D. Is exceedance higher after rain? (key question 3)

For each site with both wet and dry sample-days we compared the exceedance rate in the two conditions (Table XIV, Fig. 4). In Mexico and the United States the exceedance rate is higher after rain at the median site and at two thirds or more of sites. In Europe both medians are zero, because most sites almost never exceed the threshold, so there is no wet-dry difference to find. The pooled row is dominated by Europe's zeros and should not be read on its own. This is an association between rainfall and exceedance, not proof that rain causes it.

E. Priority ranking and dashboard

The regional ranking (questions 2 and 4) is stored in `mart_region_priority`, which holds 97 regions; those with fewer than three sampled sites are listed but not ranked. The dashboard presents the top ten regions and sites, and it states, next to the ranking, how many of the top ten are in a single country. With the current sample this ranking is largely Mexican, because all 60 GEMStat sites are in Mexico and 88% of them are persistent hotspots. We therefore present it as a ranking *within the sampled sites*, and we report that choice openly instead of hiding it. Section XIII lists what is still needed before it could guide real investment.

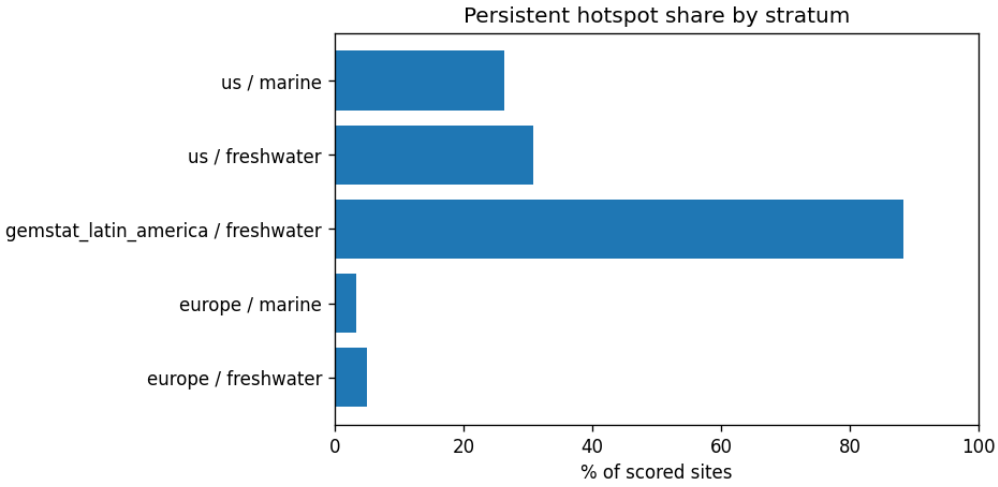


Fig. 4. Share of scored sites that are persistent hotspots. (the two Europe medians are zero, so no bars show).

XII. Challenges and Lessons

Table XV lists the problems that cost the most time and what we changed because of them. The common thread is that most of the hard problems were about *data and process*, not code.

XIII. Limitations

We list the limits of the results before the reader has to find them. Each is also stated in the README and, where relevant, on the dashboard.

A. Coverage and representativeness

- **Not global.** The results cover the United States, Europe and Mexico. Africa has no rows in the GEMStat extract and Asia has no *E. coli* rows. This reflects open-data gaps and our choice of scored indicator, not cleaner water.
- **Narrow composition.** All 60 GEMStat sites are in Mexico; Spain and Germany are 69% of the European sites; the regional ranking is therefore largely Mexican.
- **A sample, not a census.** 300 sites. Site types outside the realm mapping (about 10,455 WQP sites) and Eionet "other" water bodies (18,060 sites) are excluded, and canals and stormwater are out of scope.
- **Uneven time coverage.** Eionet ends in 2021 and the sampled GEMStat sites are effectively limited to 2015–2018, so strata are judged on different calendar years. Whether Eionet's gap is a reporting lag or a cut-off of the compilation is not established.
- **Boundaries.** French overseas territories are coded as France and so count as Europe; one of the 4,596 Natural Earth polygons is invalid and was not repaired.

B. Measurement and data

- **Laboratory limits.** About a fifth of GEMStat *E. coli* values sit at a reporting maximum and are not flagged as censored, so severity for Mexico is understated.
- **Units and censoring.** MPN is treated as CFU; censored values are substituted (half the limit for WQP; OWQ values arrive already filled, so their censoring is invisible); 1,449 WQP rows (about 11%) were dropped for units that could not be converted.
- **Weather.** Reanalysis at 0.25° is not a gauge at the site, and the local sample date is matched to a UTC weather day. The 10 mm wet threshold is an assumption.

C. Method

- **An adapted standard.** EPA's 10% rule is applied per calendar year rather than per 30-day window, and the thresholds are single-sample criteria.
- **Assumptions not yet stress-tested.** The weights, the wet threshold, the five-day minimum and the three-site minimum are judgement calls. A sensitivity check (equal weights, persistence only, wet thresholds of 5 and 20 mm) is specified in the business rules but has *not* been run on the final data, so we cannot say how much the top of the ranking depends on them.
- **Association, not causation, and no uncertainty.** The wet-dry comparison is descriptive; we report no confidence intervals or significance tests, and sites with few wet days give noisy rates.

D. Engineering

- The raw files (about 0.4 GB) are not published, so a first run downloads fresh data that may differ slightly from ours, and takes days because of the weather quota.
- The dashboard is static: its data file is regenerated and committed, not served from the warehouse.
- The Docker configuration and default passwords are for local development, not for production use.

XIV. Future Improvements

We list the improvements in the order we would do them, with the problem each one solves.

TABLE XV
MAIN CHALLENGES, OUR RESPONSE AND THE LESSON.

Challenge	What we did	Lesson
Weather quota (about 156 cells a day)	Resumable extractor that skips completed cells; a `partial` manifest that fails raw validation; weather requested only for cells that contain a sampled site	Design for provider limits from the start; make every long download restartable
WQP size and behaviour	A single national summary request timed out, so the catalogue is fetched in smaller requests; sites are sampled from it first, then results are requested only for them. Dates must be sent in the JSON body, because the endpoint ignores date filters in the query string	Read the provider's behaviour, do not assume it; a cheap spike saves days
OWQ export	A spike found a scripted export. Each file begins with a block of # header lines whose length changed between exports (7 in an early test, 6 in the final run), so readers count them instead of assuming a number	Never hard-code a layout the provider can change
Heterogeneous meaning, not just format	Mapping tables for indicators and units; a censoring rule; explicit drop reasons that make the layers reconcile	Harmonization is mostly decisions about meaning; write each one down
Data that simply is not there (Africa, Asia)	Reported as findings, in the dashboard, validation and documents. We considered scoring Asia with fecal coliform and rejected it: EPA gives no comparable recreational criterion for it, and mixing indicators would change every number	A smaller, honest scope is better than a larger, unjustified one
Review feedback on evidence	Our documents described a six-site development run and contradicted themselves (weather coverage 0–15% in one report, 100% in another), with unfinished placeholders. We regenerated everything from one final run, using scripts that *write* the evidence (profiler, EDA, benchmark, one SQL script), and added a link checker for the README	Generate evidence, do not write it; one run, one source of truth
Defaults that hide requirements	The safe default (`snapshot`) blocks downloads, and the README never said to set `live` for a first run. A fresh-clone test found it, and the README now says so	Test the instructions as a stranger would
Windows and Docker friction	PowerShell wrote UTF-16 files that GitHub showed as binary; CRLF line endings broke shell scripts; Docker needed at least 4 GB of memory. We added `.gitattributes`, UTF-8 output commands and a troubleshooting table	Document the environment problems you actually met
Duplicate DAG runs	Unpausing starts a run, and our instructions also triggered one (Section VIII)	Know what a scheduler does when you switch it on

1) **Run the sensitivity analysis.** The cheapest way to find out how much the ranking depends on our assumptions: recompute it with equal weights, with persistence only, and with wet thresholds of 5 and 20 mm, and report how many of the top ten regions change.
2) **Publish the raw snapshot.** Hosting the 0.4 GB of raw files with a download script would let anyone reproduce our exact run in minutes instead of days.
3) **Add native sources** (Hub'Eau in France, DataStream in Canada, UK Open WIMS, NMMP, LAWA in New Zealand) to fill the gaps behind the OWQ compilation, give censoring back, and add stations outside the United States and Europe. This is the direct fix for the missing Africa and Asia coverage; adding a fecal-coliform criterion would also need a defensible standard.
4) **Recover the 11% of WQP rows** dropped for unitless MPN or CFU, once the reporting volume can be confirmed per organization.
5) **Incremental loading** with a date watermark per site, so a refresh requests only new observations.
6) **Watershed-based regions** (HydroBASINS) instead of administrative regions, closer to how rehabilitation is planned.
7) **Better weather and statistics:** storm-event analysis, local-time weather, station data where available, and survival analysis for censored values instead of substitution.
8) **A live dashboard** served from PostgreSQL, a regular schedule, and alerts when a critical check fails.

XV. CONCLUSION

We built a pipeline that turns three monitoring programmes, a weather archive and a boundary file into a validated PostgreSQL warehouse, and from it a hotspot map and a regional priority ranking. Table XVI relates each objective to the evidence. The most important design choices were the immutable raw layer with checksummed manifests, the explicit drop reasons that let every row be reconciled across layers, the natural-key upsert that makes reruns safe, validation that gates the DAG, and the habit of generating evidence from code. The results are a careful but narrow picture: the United States, Europe and Mexico, over uneven years, with assumptions that have not yet been stress-tested. Within that scope, persistent hotspots are rare in European bathing waters, common in the sampled US sites, and the norm at the sampled Mexican river stations, and exceedance is higher after rain where there is any exceedance to compare. We have tried to make clear, at every step, what the results can and cannot support.

TABLE XVI
OBJECTIVES AND THE EVIDENCE FOR EACH.

#	Objective	Evidence
1	Ingest five sources in several formats	Table I; manifests with checksums; 1,362,645 raw observation rows plus 1,028,608 weather rows
2	One schema, vocabulary and unit	Section VI; reconciliation (Table IX)
3	Automated checks at three layers	42 checks, no critical failure (Table VIII)
4	Keys and relationships in PostgreSQL	Section V; foreign keys and uniqueness checked in the database
5	Airflow in Docker	Section VIII; fresh-clone reproduction (Section IX)
6	Hotspots and regional ranking	Tables XIII; dashboard; 97 regions in the priority mart
7	Rain and exceedance	Table XIV: higher after rain in the United States and Mexico

APPENDIX A
REPRODUCING THE RESULTS

The commands below assume a machine with Docker; the repository README has the full instructions, including Windows notes. On a machine with an empty `data/raw/` folder, first set `PIPELINE_MODE=live` in `.env` (the first download takes days because of the weather quota); with the raw files in place, the default `snapshot` mode runs offline. Unpausing the DAG starts the first run by itself; do not also trigger it.

```
git clone https://github.com/labillescharlene21-lab/project.
    ↪ git
cd project
cp .env.example .env
docker compose up -d --build --wait
docker compose run --rm pipeline python -m pytest -q
docker compose exec airflow-scheduler airflow dags unpause
    ↪ wq_hotspot_pipeline
```

Reports and evidence are regenerated from the final data with:

```
docker compose run --rm -T pipeline python -m src.analysis.
    ↪ profile_sources
docker compose run --rm -T pipeline python -m src.analysis.
    ↪ eda
docker compose run --rm -T pipeline python -m src.analysis.
    ↪ export_site
docker compose run --rm -T pipeline python -m src.analysis.
    ↪ benchmark_formats
docker compose exec -T postgres psql -U wq_user -d
    ↪ water_quality < sql/evidence.sql
```

APPENDIX B
VALIDATION CHECK CATALOGUE

TABLE XVII
ALL 22 VALIDATION CHECKS.

Stage	Check	Severity	What it tests
Raw	`manifest_status_success`	critical	The extraction finished (not `partial` or `failed`)
Raw	`files_exist_non_empty`	critical	Every file in the manifest exists and is not empty
Raw	`checksum_matches`	critical	Each file's SHA-256 still matches the manifest
Raw	`row_count_matches`	critical	Data rows on disk equal the manifest count
Raw	`required_columns_present`	critical	Data files contain every column of the source's schema file
Raw	`json_shape`	critical	Weather files have the required keys; every daily series is as long as the date series
Staging	`schema_matches`	critical	Columns and types match the staging schema
Staging	`keys_not_null`	critical	Columns declared non-nullable contain no nulls
Staging	`obs_key_unique`	critical	No observation key appears twice
Staging	`accepted_values`	critical	Indicator, source, censor direction and realm use allowed values only
Staging	`value_range`	warning	No negative values; values above 10^6 flagged
Staging	`date_in_period`	critical	Every sample date is in the period and matches its year partition
Staging	`coordinates_valid`	critical	Latitude in $[-90, 90]$ and longitude in $[-180, 180]$
Staging	`observation_site_exists`	critical	Every observation's site exists in the sites table
Staging	`stratum_coverage`	warning	Each stratum and realm has enough sampled sites
Curated	`tables_present`	critical	All 11 warehouse tables exist
Curated	`referential_integrity`	critical	Every foreign key has no orphan rows (read from the database catalogue)
Curated	`pk_unique_in_db`	critical	No duplicated natural-key values in any table
Curated	`reconciliation`	critical	Raw rows minus drops equal staging rows; staging rows equal warehouse rows
Curated	`weather_coverage`	warning	At least 90% of observations have a rain value
Curated	`mart_rank_consistency`	critical	Ranks run 1..n without gaps; unranked regions have no rank
Curated	`hotspot_share_sane`	warning	The hotspot share is strictly between 0% and 100%

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